

BEFORE

Before the 17th century, religion was practically unchallenged as a framework for understanding the world, but the scientific revolution offered alternative explanations for our existence.



GALILEO FACES THE INQUISITION

GALILEO AND THE INQUISITION

In 1632, Italian scientist Galileo Galilei asserted his belief that Earth revolved around the Sun—a theory first put forward a century earlier—in his book *Dialogues on the Two Chief World Systems*. The Catholic Church banned it, and the Inquisition (a religious tribunal) **◀ 196** forced

Galileo (above) to denounce his own work as heretical. He was placed under house arrest, but during his confinement he wrote *Discourses Upon Two New Sciences*, which later inspired Isaac Newton **◀ 269** to scientific discoveries that further challenged Biblical doctrine.

SOCIETIES AND ACADEMIES

During the 17th century the emergence of **scientific societies** such as the Royal Society in Britain and the Académie des Sciences in France provided forums for debate and encouraged the **formalization of the scientific method**. The societies also gave science respectability, though it continued to face opposition from the Church.

NEWTON'S GOD

Isaac Newton's scientific discoveries were criticized because it was thought that they had **demystified the world**, but Newton was



MECHANICAL MODEL OF THE SOLAR SYSTEM

devoutly religious: he did not present **science as an alternative to God**, and nothing in his work intentionally excluded God from the universe he described. Newton believed that God was the force behind the natural systems his science had sought to explain, with the universe as a clockwork machine (above) created by God.

THE ENLIGHTENMENT

Newton's discoveries caused a **revolution in physics** and inspired a new intellectual and philosophical movement known as the Enlightenment **◀ 270–71**, whose thinkers based their work on **reason and evidence**, rather than superstition and accepted beliefs.

“Was it through his grandfather or his grandmother that he claimed his **descent from a monkey?**”

BISHOP WILBERFORCE, ADDRESSING THOMAS HUXLEY AT A MEETING OF THE BRITISH ASSOCIATION FOR THE ADVANCEMENT OF SCIENCE, 1860

On June 30, 1860 science clashed with religion on a stage at Oxford University's Museum Library. Weighing in on one side of the debate were Bishop Samuel Wilberforce and the Anglican Church; on the other was English biologist T. H. Huxley, armed with a copy of *On the Origin of Species*, published by Charles Darwin, the British naturalist, the year before (see p.341). Darwin's book, which advanced the theory that all creatures had evolved through natural selection, critically undermined the literal interpretation of the Bible, which held that God created the world, including humans, during a period of six days around 6,000 years ago. But religion was more than ready to defend

Heated debate

Cartoons from the magazine *Vanity Fair* show Bishop Samuel Wilberforce (left) and T. H. Huxley. Their debate became known as “the moment that science clashed with religion.”



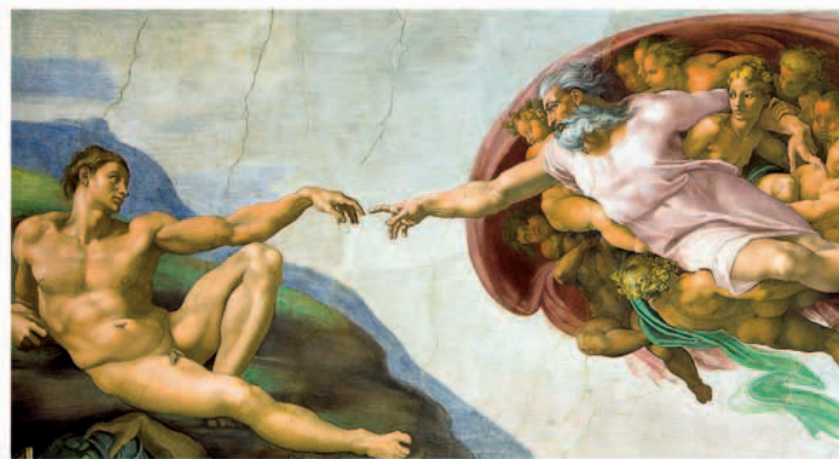
Science vs. God

In 1859 Charles Darwin published *On The Origin of Species by Means of Natural Selection*, in which the naturalist put forward his theory of evolution. This sparked a public debate over the roles of religion and science in providing the means by which humans understood the world around them.

its position. Bishop Wilberforce attacked Huxley over a perceived lack of evidence, while Huxley quipped that he would rather be descended from a monkey than a bishop.

Hard evidence

This challenge to previously accepted ideas was not a deliberate attack on religion by science, but came simply from the fact that discoveries were being made that were at odds with Christian doctrine. As a Christian himself, Darwin found the situation very difficult and was reluctant to publish his theory for fear of the public's reaction, ruefully referring to himself as “the Devil's Chaplain.” Philosophers had been challenging the literal interpretation of the Bible since the Enlightenment (see p.270–71), but in the late 18th and early 19th



centuries, scientific evidence added a new dimension to the debate. From 1785 to 1835, four scientists—James Hutton, Georges Cuvier, William Smith, and Sir Charles Lyell—provided

The Creation of Adam

Michelangelo's depiction of God creating Man adorns the ceiling of the Sistine Chapel in Rome. It is part of a series of panels depicting stories from the book of Genesis, the first book of the Bible.